CERTIFIED CONSTRAINTS ON COUNTEREXAMPLES TO AGRAWAL'S CONJECTURE AT $r = 5$

THE UNICO/NOUS PIPELINE

Abstract. This note collects the part of our structural study of Agrawal's conjecture at $r = 5$ that is mechanically verified. Every statement called a theorem below is proved in Lean 4 over Mathlib, with no `sorry`, no `admit` and no added axioms; the corresponding declaration name is given with each statement. Every computational claim carries a replayable certificate with its detection criterion embedded and SHA-256 manifests. We then pose, without any claim of proof, the two open problems that our working material identifies as central. Nothing in this note asserts the conjecture.

1. Setting

Agrawal's conjecture (arising from the AKS primality test [1]; see also [2, 3, 4]) states: if r is prime, $r \nmid n$, and $(X-1)^n \equiv X^n - 1 \pmod{n,\, X^r - 1}$, then n is prime or $n^2 \equiv 1 \pmod{r}$. Our study concerns $r = 5$. This note isolates its mechanically certified core; the statements are elementary, and their interest lies in the role they play in the wider study and in the fact that a proof assistant has checked every one of them.

In Sections 2–4, p is an odd prime; in Section 5, q denotes an arbitrary prime factor of n, with $q = 2$ allowed. Let

$$R_p \;=\; (\mathbf{Z}/p)[X]\,/\,(X^2 - X - 1), \qquad \varepsilon \;=\; \text{the class of } X,$$

so that $\varepsilon^2 = \varepsilon + 1$, and set $\sqrt{5} := 2\varepsilon - 1$, which satisfies $(\sqrt{5})^2 = 5$ in R_p (`GoldenRing`, `eps_sq`, `sqrt5_sq`). Let F_n, L_n denote the Fibonacci and Lucas numbers, with $L_n = F_{n+1} + F_{n-1}$ (`lucas`). Define

$$A_n = \begin{cases} L_n, & n \text{ even}, \\ F_n, & n \text{ odd}, \end{cases} \qquad H_n = \gcd\bigl(A_n,\, 5^{\,n-1} + 1\bigr),$$

and, dually, $B_n = F_n$ (n even), L_n (n odd), with $J_n = \gcd\bigl(B_n,\, 5^{\,n-1} - 1\bigr)$.

An autonomous multi-model pipeline (Claude Fable 5, Claude Opus 4.8, GPT-5.6-Sol, with an independent external review pass), orchestrated by Daniele Cappello, who carried the messages, audited the artifacts and made no mathematical contribution. Repository with the Lean sources and all computational certificates: `https://github.com/Solarys431/agrawal-r5`.

2. The golden Frobenius and the inertia of J_n

Theorem 2.1 (`golden_frobenius`). *Assume* 5 *is not a square in* $\mathbf{Z}/p$ *and* $p \neq 2$. *Then* $\varepsilon^p = 1 - \varepsilon$ *in* R_p.

Certified proof sketch. By Euler's criterion (`euler_nonsquare`) we have $5^{(p-1)/2} = -1$ in $\mathbf{Z}/p$, hence

$$(\sqrt{5})^p = (5)^{(p-1)/2}\sqrt{5} = -\sqrt{5} \qquad (\texttt{sqrt5\_pow\_p}).$$

Expanding $(2\varepsilon - 1)^p$ by the Frobenius endomorphism and cancelling the unit 2 gives the claim. □

Corollary 2.2 (`golden_pow_p_succ`). *Under the same hypotheses,* $\varepsilon^{p+1} = -1$ *in* R_p.

Theorem 2.3 (`inertia_J`). *Assume* 5 *is not a square in* $\mathbf{Z}/p$ *and* $p \neq 2$. *Then for no* $n \geq 1$ *do* $\varepsilon^{2n} = 1$ *in* R_p *and* $5^{n-1} = 1$ *in* $\mathbf{Z}/p$ *hold simultaneously.*

Certified proof sketch. If n is odd, raising $\varepsilon^{2n} = 1$ to the power $(p+1)/2$ and comparing with $(\varepsilon^{p+1})^n = (-1)^n = -1$ gives $1 = -1$, absurd for $p \neq 2$. If n is even, $5 = 5^n = (5^{n/2})^2$ exhibits 5 as a square, against the hypothesis. □

Via the identity $\varepsilon^{n+1} = F_{n+1}\varepsilon + F_n$ in R_p (`golden_pow_fib` and its predecessor form `golden_pow_pred`) one obtains the divisibility forms, in which the hypothesis is purely arithmetic by quadratic reciprocity (`not_isSquare_five`):

Theorem 2.4 (`inertia_J_fib_mod5`, `inertia_J_lucas_mod5`). *Let* $p \equiv \pm 2 \pmod 5$, $p \neq 2$. *For even* $n \geq 2$, p *does not divide* $\gcd(F_n, 5^{n-1} - 1)$; *for odd* $n \geq 1$, p *does not divide* $\gcd(L_n, 5^{n-1} - 1)$. *Equivalently, no prime inert in* $\mathbf{Q}(\sqrt{5})$ *divides any* J_n.

3. The canonical support witness

Theorem 3.1 (`support_witness`). *Let* $p \equiv 2 \pmod 5$, $p \neq 2$, *and put* $n_0 = (p+1)/2$. *Then* $n_0 \equiv 4 \pmod 5$, $p \mid 5^{n_0 - 1} + 1$, *and* p *divides* L_{n_0} *if* n_0 *is even,* F_{n_0} *if* n_0 *is odd. In particular* $p \mid H_{n_0}$, *with an index in the class* $4 \bmod 5$.

4. The golden–cyclotomic entanglement

Theorem 4.1 (`entanglement`). *In any commutative ring, if* $\zeta^4 + \zeta^3 + \zeta^2 + \zeta + 1 = 0$, *then*

$$\zeta^3\,(1 + \zeta + \zeta^4)^2\,(\zeta - 1)^4 \;=\; 5.$$

In particular this holds in $\mathbf{Z}[\zeta_5]$ *with* $\varepsilon = 1 + \zeta + \zeta^4$ *the image of the golden unit.*

The proof is a single `linear_combination` with an explicit cofactor; $\zeta^5 = 1$ follows from the same hypothesis (`zeta_pow_five`).

5. The Fermat-shadow glue

Lemma 5.1 (`mul_dvd_gcd_mul`). *If $x \mid M$ and $y \mid M$, then $xy \mid \gcd(x, y)\, M$.*

Theorem 5.2 (`fermat_shadow`). *Let n be squarefree with $5 \nmid n$, and suppose that for every prime $q \mid n$*

$$5^{\,n/q-1} \equiv 1 \pmod{q}. \tag{$*$}$$

Then $n \mid 5^{\,n-1} - 1$; that is, n is a Fermat pseudoprime to base 5.

Remark 5.3. Hypothesis ($*$) is the local norm constraint of the wider study; in Theorem 5.2 it is an explicit hypothesis. The next theorem removes it: the Agrawal congruence itself forces base-5 pseudoprimality, unconditionally.

Theorem 5.4 (`agrawal_fermat_shadow`). *Let n be squarefree with $5 \nmid n$, and suppose the Agrawal congruence at $r = 5$ holds: $(X - 1)^n \equiv X^n - 1 \pmod{n, X^5 - 1}$. Then $n \mid 5^{\,n-1} - 1$. In particular every squarefree counterexample to Agrawal's conjecture at $r = 5$ prime to 10 is a Fermat pseudoprime to base 5.*

Certified proof sketch. Reduce the congruence mod a prime $q \mid n$; it becomes a polynomial identity in $(\mathbf{Z}/q)[X]$, which can be evaluated at the four elements z^j $(j = 1, \ldots, 4)$ of the quotient ring $(\mathbf{Z}/q)[X]/\Phi_5$ (not a field in general), where $z^5 = 1$ kills the modulus. Multiplying the four evaluated identities and using the explicit-cofactor identity $(z-1)(z^2-1)(z^3-1)(z^4-1) = 5$ (`prod_pow_sub_one`) gives $5^{\,n} = 5$ in the quotient ring; injectivity of the constant embedding descends this equality to $\mathbf{Z}/q$, hence $5^{\,n-1} = 1$ there (`agrawal_local`), and the squarefree gluing finishes. No Frobenius descent and no norm maps are needed. □

6. The two-adic jaw

The bridge constrains a squarefree solution of the Agrawal congruence only through the multiplicative order of 5. On the primes that are inert in $\mathbf{Q}(\sqrt{5})$ that constraint is rigid at the prime 2.

Theorem 6.1 (`agrawal_two_adic_jaw`). *Let n be squarefree with $5 \nmid n$ and suppose the Agrawal congruence at $r = 5$ holds. Let $q \mid n$ be an odd prime with $q \equiv \pm 2 \pmod{5}$. Then*

$$v_2(q - 1) \;\le\; v_2(n - 1).$$

Certified proof sketch. By Theorem 5.4, $\mathrm{ord}_q(5)$ divides $n-1$. Since $q \equiv \pm 2 \pmod{5}$, quadratic reciprocity gives $(5/q) = -1$ (`not_isSquare_five`), so $5^{(q-1)/2} = -1$ (`euler_nonsquare`) and the order does *not* divide $(q-1)/2$. A divisor of $2^a u$ with u odd that fails to divide $2^{a-1}u$ is 2-adically saturated (`pow_two_dvd_of_not_dvd_half`), whence $v_2(\mathrm{ord}_q 5) = v_2(q-1)$, and the claim follows; the local form is `two_adic_jaw_local`. □

Corollary 6.2 (`agrawal_inert_mod_four`). *Under the same hypotheses, if $n \equiv 3 \pmod 4$ then every prime factor of n that is inert in $\mathbf{Q}(\sqrt{5})$ satisfies $q \equiv 3 \pmod 4$.*

Remark 6.3. The split primes behave in the opposite way: for $q \equiv \pm 1 \pmod 5$ one has $(5/q) = +1$, hence $\mathrm{ord}_q(5) \mid (q-1)/2$ and $v_2(\mathrm{ord}_q 5) < v_2(q-1)$, so no constraint is transmitted. Both statements above are unconditional.

7. The bounded order

The bridge and the jaw both read the congruence through the multiplicative order of an element. The element that carries the congruence itself is $\zeta - 1$, where ζ is the class of X in $T = \mathbf{F}_p[X]/\Phi_5$. A priori its order is only known to divide $|T^\times| = p^4 - 1$. It divides much less.

Theorem 7.1 (`order_bounded`). *Let p be a prime with $p \equiv \pm 2 \pmod 5$, so that Φ_5 stays irreducible modulo p, and let ζ be the class of X in $T = \mathbf{F}_p[X]/\Phi_5$. Then*

$$(\zeta - 1)^{10p^2} = (\zeta - 1)^{10}.$$

Equivalently, the multiplicative order of $\zeta - 1$ divides $10(p^2 - 1)$.

Certified proof sketch. Write $u = \zeta - 1$. In characteristic p the Frobenius is additive, so $u^p = \zeta^p - 1$, and since $\zeta^5 = 1$ the exponent only matters modulo 5 (`zeta5_pow_mod`), giving $u^p = \zeta^{p \bmod 5} - 1$ (`zeta_sub_one_pow_p`). Applying the Frobenius once more multiplies that exponent by p again, and here the two inert classes merge: $2^2 \equiv 3^2 \equiv 4 \pmod 5$, so in both cases $u^{p^2} = \zeta^4 - 1$ (`zeta_sub_one_pow_sq`). Finally $\zeta^4 = \zeta^{-1}$ turns that into

$$u^{p^2} = \zeta^{-1} - 1 = -(\zeta - 1)\zeta^{-1} = -\zeta^4 u$$

(`order_bound_relation`): the square of the Frobenius merely scales u by $-\zeta^4$. Raising to the tenth power kills the scalar, since $(-1)^{10} = 1$ and $\zeta^{40} = (\zeta^5)^8 = 1$. □

Remark 7.2. The statement is phrased as $u^{10p^2} = u^{10}$ rather than as a divisibility of the order, so that no invertibility of u is needed. It is the form actually used downstream: the factor 10 is what forces the modulus 80 to appear in the congruence conditions on the prime factors of a putative counterexample, and it is what makes an exact-order computation factor a ten-digit integer instead of a twenty-digit one.

8. An elementary box

The following identity is used to confine the largest prime factor of a three-factor solution to an interval. It is stated over any commutative ring and proved by expansion.

Theorem 8.1 (`prod_pairs_sub_prod_squares`). *In any commutative ring, for all p, q, r,*

$$2\left[(pq-1)(pr-1)(qr-1)-(p^2-1)(q^2-1)(r^2-1)\right] = (p^2-1)(q-r)^2+(q^2-1)(p-r)^2+(r^2-1)(p-q)^2.$$

In particular the left-hand side is non-negative whenever $p, q, r \geq 1$ (`prod_pairs_ge_prod_squares`).

Combined with `lt_two_mul_of_sq_le`, which turns $r^2 \leq 2I(pq)$ with $q < r$ into $r < 2Ip$, this bounds the third prime by an explicit multiple of the smallest one.

9. The Lenstra–Pomerance proposition

The statement below is the one on which the search for counterexamples at $r = 5$ rests. It appears in the AIM notes of Lenstra and Pomerance [2, pp. 30–32] and is quoted throughout the subsequent literature; to our knowledge it had not been machine-checked before.

Mathlib does not currently contain Carmichael numbers or Korselt's criterion, so those had to be introduced: `IsCarmichael` and `IsLucasCarmichael` package the two divisibility conditions used below, while `IsCarmichael.fermatPsp` Korselt's criterion proper, that a Carmichael number is a Fermat pseudoprime to every coprime base is proved for completeness and is *not* used in the proof of Theorem 9.1, which needs only the divisibility conditions themselves.

Theorem 9.1 (`lenstra_proposition_card`). *Let n be squarefree, suppose the number k of its prime factors satisfies $k \equiv 1 \pmod 4$, suppose every prime $p \mid n$ satisfies $p \equiv 3 \pmod{80}$, and suppose Korselt's two conditions hold: $(p-1) \mid (n-1)$ and $(p+1) \mid (n+1)$ for every $p \mid n$. Then*

$$(X-1)^n \equiv X^n - 1 \pmod{n,\ X^5 - 1} \qquad \textit{and} \qquad n^2 \not\equiv 1 \pmod 5.$$

These are the hypotheses of the original statement. That $n \equiv 3 \pmod{80}$ follows from them is itself proved (`mod_eighty_of_card`): for squarefree n the product of the prime factors is n, each factor is 3 modulo 80, and $3^4 = 81 \equiv 1 \pmod{80}$ reduces the exponent to the class of k modulo 4. A variant assuming $n \equiv 3 \pmod{80}$ directly is available as `lenstra_proposition`.

The second conclusion is what makes such an n, *if composite*, a genuine counterexample: Agrawal's conjecture allows the escape $n^2 \equiv 1 \pmod r$, and here it is closed (`sq_not_one_mod_five`). The proviso matters: primes satisfy the hypotheses trivially $n = 83$ is one and are of course not counterexamples. Only a composite witness would be one, and none is known.

Certified proof sketch. The route is the one of the original proof, which already contains the identity $(\zeta_5 - 1)^{p^2} = -\zeta_5^{-1}(\zeta_5 - 1)$, the resulting bound on the order, and the reduction to $n \equiv p$; we repackage it through a single observation: *the three conditions imposed on n are satisfied by p itself*, since $p \equiv 1 \pmod{p-1}$, $p \equiv -1 \pmod{p+1}$ and $p \equiv 3 \pmod{80}$. Hence

$$n \equiv p \pmod{\operatorname{lcm}(p-1,\ p+1,\ 80)}, \qquad \operatorname{lcm}(p-1,\ p+1,\ 80) = 10(p^2-1)$$

(`lcm_three_eq`, `sub_dvd_of_korselt`), and that modulus is exactly the one bounding the order of $\zeta - 1$ by Theorem 7.1. Therefore $(\zeta-1)^n = (\zeta-1)^p$, and the Frobenius gives $(\zeta-1)^p = \zeta^p - 1 = \zeta^3 - 1 = \zeta^n - 1$, which is the congruence in the cyclotomic component (`lenstra_local`). On the component $X \mapsto 1$ both sides vanish, and since $X^5 - 1 = (X-1)\Phi_5$ with the two factors coprime for $p \neq 5$, the two pieces glue to the congruence modulo p (`agrawal_mod_p`, using `dvd_of_dvd_both`). Finally, for squarefree n a congruence holding modulo every prime factor holds modulo n: dividing by the monic $X^5 - 1$ over $\mathbf{Z}$, every coefficient of the remainder is divisible by each $p \mid n$, hence by n (`congruence_of_local`). □

Remark 9.2. Nothing here asserts that a composite such n exists. A composite number satisfying the hypotheses would be simultaneously a Carmichael and a Lucas–Carmichael number; no such integer is known, and the question is open since Pomerance raised it in 1984 in connection with the Baillie–PSW primality test. Note also that the proposition gives a *sufficient* condition for producing a counterexample, not a necessary one: it does not say where all counterexamples must lie. It certifies the tool used to look for them; it does not decide Agrawal's conjecture in either direction.

10. Computational certificates

Independently of the kernel-checked statements above, the repository carries replayable certificates, each embedding its own detection criterion and reconstruction data, with SHA-256 manifests:

(1) seven *certified-empty fibres* of a three-factor sieve associated with the study (pairs $(13, 37)$, $(17, 113)$, $(13, 277)$, $(23, 167)$, $(7, 823)$, $(83, 347)$, $(463, 1327)$): for each pair, the finitely many levels required by the certificate are integrally factored, all prime factors are proven prime, and no prime in the certificate's detector classes survives;
(2) a self-certifying census of H_n for all $n \equiv 4 \pmod 5$, $n \leq 10^5$: 9,725 nontrivial values, all factorizations proven, and *no* prime factor $\equiv \pm 1 \pmod 5$ occurs.

These are certificates of finite computations; they prove exactly what they state and nothing beyond, and both replay from the repository alone. Separately, the repository freezes as *hash-only provenance* (hashes, not artifacts) a SHA-256 index of the working material of the study, including a prime-by-prime corpus up to 10^9; nothing in this note relies on it.

11. Two open problems

We pose the two statements that our working material identifies as the central open problems of the $r = 5$ study. We claim no proof and no partial proof here.

Conjecture 11.1 (golden inertia)**.** *For every* $n \equiv 4 \pmod 5$*, every odd prime factor of* H_n *distinct from* 5 *is* $\equiv \pm 2 \pmod 5$*.*

Theorem 3.1 shows that every prime $p \equiv 2 \pmod 5$ does occur in some H_n with $n \equiv 4 \pmod 5$; the census in item (2) above verifies Conjecture 11.1 for all $n \leq 10^5$.

Conjecture 11.2 (fibre emptiness)**.** *The pattern of item (1) above is general: for every admissible pair, the corresponding fibre is empty.*

12. Formalization data

Twenty-two modules, 2312 source lines, 126 named declarations; no `sorry`, no `admit`, no axioms beyond Mathlib's. Mathlib is pinned to a fixed upstream commit; a clean clone builds with `lake exe cache get && lake build`. The table mapping each statement of this note to its declaration and file is in the repository `README`.

References

[1] M. Agrawal, N. Kayal, and N. Saxena, *PRIMES is in P*, Ann. of Math. 160 (2004), 781–793.

[2] H. W. Lenstra, Jr. and C. Pomerance, *Remarks on Agrawal's conjecture*, AIM note, 2003.

[3] H. W. Lenstra, Jr. and C. Pomerance, *Primality testing with Gaussian periods*, J. Eur. Math. Soc. 21 (2019), 1229–1269.

[4] T. Váňa, *Agrawal's conjecture and Carmichael numbers*, Student Scientific Conference, Comenius University Bratislava, 2009.